10. ESTIMATION OF AMINO ACID BY SORENSEN'S FORMOL TITRATION METHOD

Aim:

To estimate the amount of amino acid present in100ml the given solution by formol titration

Principle:

The carboxyl group of amino acid cannot be accurately measured by titration with alkali because it reacts with a basic amino acid group to form zwitter ions of amphoteric nature at neutral pH. They may not be decomposed at the end point. But in the presence of formaldehyde with alkali, the amino acid forms an amino acid formol complex. The complex is acidic because of the basic character of amino acid group is suppressed and hence it can be titrated with alkali using phenolphthalein as indicator.

Reagents:

1. **Stock glycine solution:** 7g of glycine is made up to 100ml with distilled water.
2. 630mg of oxalic acid is made up to 100ml with distilled water.
3. Sodium hydroxide solution (0.1N): 4g of made up to 1000ml with distilled water.
4. 40% Formaldehyde
5. Phenolphthalein indicator.

Procedure:

Standardisation of sodium hydroxide: 0.1N of oxalic acid was prepared (0.63g of oxalic acid was dissolved in 100 ml of water). 10ml of oxalic acid was pipetted out into a conical flask and titrated against sodium hydroxide using phenolphthalein as indicator. End point is the appearance of pale pink colour. The experiment was repeated for concordant values.

Estimation of amino acid: 10 ml of formaldehyde and 10ml of distilled water are pipetted out into a conical flask. This is taken as blank. This is titrated against standard sodium hydroxide using phenolphthalein as indicator. A permanent pink colour obtained is the end point. Burette reading is noted and the titration is repeated for concordant values.

10ml of 0.1 N glycine, 10ml of formaldehyde and 10ml of distilled water are pipette out into a conical flask. The solution is allowed to stand for ten minutes to complete the reaction. The above mixture is titrated against sodium hydroxide using phenolphthalein as indicator. A pale pink colour was obtained. End point was noted and the experiment was repeated for concordant values. Similarly the titration is repeated with the unknown glycine and the amount of glycine in the unknown sample is calculated.

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